

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 38

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Renewable: re-forms naturally relatively quickly so use does not necessarily reduce future availability (solar, wind, tidal, geothermal, biofuel) (1) • Depletable: use can reduce future availability - all non-renewables (fossil fuels, uranium) plus renewables exploited unsustainably (1) • Wood / biofuels are renewable but depletable if forests are felled faster than they regrow (1)	3
01.2	<i>1 mark for each point.</i> • Other factors (technology, location, cost) restrict how much can be harnessed (1) • Examples: coal too deep to mine; winds high above the ground / far offshore where turbines cannot be placed; waves in deep water where equipment cannot be anchored; low-intensity sunlight that cannot produce high temperatures (1) • New technologies or higher prices may make these usable in future (1)	3
01.3	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Wind: strong reliable winds - shallow seas, open plains, uplands; low land-use conflict (not near urban areas, ecologically sensitive or scenic areas); near the grid (1) • HEP: high reliable rainfall, large catchment, narrow valley for a small dam with a large basin, impermeable stable bedrock (1) • Solar: high light intensity, low cloud cover (1) • Wave: coasts with strong reliable prevailing winds and a long fetch (1) • Tidal: large tidal range, coastal features (estuaries, channels) that focus the flow (1) • Geothermal: hot rocks near the surface / recent volcanic activity (1) • Biofuels: nearby farmland / forest / urban waste to avoid transporting bulky low-density fuel (1)	4

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-177; AQA spec 3.3.2.	
02.1	<i>1 mark for each point.</i> - Clay particles are very small and carry electrostatic charges on their surfaces so they repel each other and stay suspended (1) - Flocculants (aluminium sulfate / alum or polyelectrolytes) are added and mixed to neutralise the charges (1) - The particles clump together (flocs) and settle in the clarifier tank (1)	3
02.2	<i>1 mark for each point.</i> - Increases dissolved oxygen and removes hydrogen sulfide (bad-egg smell) from anaerobic sources (1) - Oxidises some dissolved metals (eg iron) so they become insoluble and can be removed; improves taste (1)	2
02.3	<i>1 mark for each point.</i> - Peaty water contains organic matter which reacts with chlorine to produce toxic substances (eg trihalomethanes) (1) - Ozone / UV kill pathogens without forming these by-products (though chlorine / chloramine keeps water sterile during distribution) (1)	2
02.4	<i>1 mark for each point.</i> - Ion exchange: toxic ions adsorb onto polymer resin beads or zeolite (also used to soften water by removing calcium / magnesium) (1) - Reverse osmosis / distillation (desalination) - expensive, energy-intensive (1) - Activated carbon: organic molecules causing odour / taste and pesticides adsorb onto the high-surface-area carbon particles (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 126-128; AQA spec 3.2.2.4.	
03.1	<i>1 mark for each point.</i> - Use a calibrated sound meter (dB(A)) at fixed distances (eg 10, 50, 100, 250, 500 m) from the track behind each barrier and at a control site with no barrier (1) - Standardise the noise source: same cars / same race or a standard test noise; same time of day, wind speed / direction, weather; same height of meter above ground (1) - Take repeated readings at each distance and calculate means (and standard deviation) to assess variability (1) - Use a measure such as LA10 or Leq over a set period rather than a single peak reading (1) - Compare reduction in dB relative to the control at each distance; plot noise level against distance for each barrier; statistical test (t-test) to assess significance (1)	5
03.2	<i>1 mark.</i> Wind carries sound / affects how far it travels so different wind directions would change readings independently of the barrier (1)	1
03.3	<i>1 mark for each point.</i> - LA10 is the noise level exceeded for 10% of the period - a measure of the noisiest periods (1) - LA90 is the level exceeded for 90% of the period - a measure of the quietest / background level (1)	2

Question	Marking guidance	Marks
03.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Double / triple glazing (1) • Acoustic insulation in walls / roof (1) • Bedrooms on the side facing away from the track (1) • Dense hedges / trees and earth mounds in gardens (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 293-295, 418; AQA spec 3.4.3.2.13.	
04.1	<i>1 mark for each point.</i> • Net held with its straight edge on the river bed facing the current (open to the flow) (1) • The bed in front is disturbed by kicking / scraping a boot for a standard number of kicks and time (eg 30 s) (1) • Same substrate type, water depth and flow rate at both sites; same net mesh; repeat samples; invertebrates identified with a key and counted (1)	3
04.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Difficult to standardise the vigour / area kicked (1) • Buried organisms or those fixed to stones are not dislodged (1) • Some organisms swim and avoid the net (1) • Flow rate varies and affects capture (1) • Fragile organisms destroyed (1)	3
04.3	<i>1 mark for each point.</i> • A fixed area of bed is enclosed by a frame so the sampled area is standardised; side nets reduce escapes (1) • But only works in fast enough flow and is hard to set on rough substrates / deep water (1)	2
04.4	<i>1 mark for each point.</i> • Pollutant / oxygen levels fluctuate (eg with flow, time of day) so one reading may miss a pollution event (1) • Sensitive taxa (stonefly, mayfly larvae) disappear and tolerant ones (midge larvae, hoglouse) remain, integrating conditions over weeks / months (biotic index) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 395-396, 415-416; AQA spec 3.7.2.1.	
05.1	<i>1 mark for each point.</i> • Predicting the route / movement a pollutant will take through the environment (based on its properties and features of the environment) (1) • To assess where it will become concentrated / which locations and organisms are most at risk, so monitoring can focus on those sites (1)	2
05.2	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Sea current velocity and direction (1) • Properties of the radionuclides: half-life, solubility, tendency to adsorb onto sediments / bioaccumulate (1) • Food chain pathways, eg uptake by seaweed, shellfish, fish eaten by people (1) • Geology / hydrology of the coast; sediment deposition areas (mud flats) (1) • Human activities in the area: fishing, seaweed collection, beach use (1)	3

Question	Marking guidance	Marks
05.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> - Identify the group of people whose lifestyle gives them the highest exposure (eg people who eat a lot of locally caught shellfish / fish, live near the mud flats, collect seaweed) (1) - Monitor their exposure / the concentration of radioactive material in the foods they eat or their body (1) - If their exposure / risk is acceptably low, everyone else is assumed to be at even lower risk; if not, emissions are controlled (1) - CGM assesses exposure before health effects occur (not used to detect damage) (1)	3
05.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Seawater (1) - Sediment / mud (1) - Seaweed (1) - Shellfish / molluscs / fish (1) - Milk, vegetables or meat from nearby farms (1) - Atmospheric dust (1) - Soil (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 243, 303-304; AQA spec 3.4.3.1.	
06.1	<i>1 mark for each point.</i> - CO₂ rises from about 412 ppm in January to a peak of 416 ppm in May, falls to a minimum of 407 ppm in September, then rises again (1) - During spring / summer, photosynthesis by plants (growing season) removes CO₂ faster than respiration releases it, so the concentration falls (1) - In autumn / winter photosynthesis is reduced (leaf fall) but respiration and decomposition continue, so CO₂ rises (1) - The atmosphere is in dynamic equilibrium: the concentration fluctuates around a mean; more land / vegetation in the northern hemisphere makes the cycle large (1)	4
06.2	<i>1 mark for each point.</i> - Natural processes that add gases (respiration, combustion, volcanoes, denitrification) are balanced by processes that remove them (photosynthesis, dissolving, nitrogen fixation) (1) - So the average composition stays the same over long timescales although the rates vary and concentrations fluctuate around a mean (1)	2
06.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Absorbs harmful UV / solar wind radiation in the stratosphere (ozone layer) (1) - Greenhouse gases delay the escape of infrared, keeping the surface warm (33 °C warmer than without) (1) - Atmospheric pressure allows liquid water to exist (1) - Distributes heat from the tropics to higher latitudes by winds (1) - Transports water vapour to areas that would otherwise be dry (1)	2
06.4	<i>1 mark for each point.</i> - Troposphere (1) - Stratosphere (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-89; AQA spec 3.2.1.1.	

Question	Marking guidance	Marks
07.1	<i>A correct final answer (with unit where asked) scores full marks. If the answer is wrong, award 1 mark for each correct step of working shown.</i> • $(2900 - 1400) / 1400 \times 100$ (1) • = 107% (accept 107.1) (1)	2
07.2	<i>1 mark for each point.</i> • Large decrease from 24 per year (1975-79) to 2 per year (2010-14) (1) • Steepest fall between the late 1970s and 1980s; slower decline / levelling off after 1995 (1)	2
07.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Double-hulled tankers required for all large tankers (1) • Improved navigation (GPS, AIS) and offshore shipping routes (1) • Inert gas systems reducing explosions (1) • Twin engines / rudders reducing groundings after mechanical failure (1) • Stricter regulation, inspection and polluter-pays liability (eg after Exxon Valdez) (1) • Separate ballast tanks / crude oil washing (1)	3
07.4	<i>1 mark for each point.</i> • Most oil pollution comes from dispersed / diffuse sources not tanker accidents: waste lubricating oil, industrial oil in drains, road runoff (1) • Routine operations: small spills, pipeline leaks, storage tank leakage, oil rig accidents (eg Deepwater Horizon) (1) • Growth in oil use and transport means more small releases even if large accidents are rarer (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 258-262; AQA spec 3.4.3.2.7.	
08.1	<i>1 mark.</i> • The length of time a pollutant remains in the environment before it is broken down / degraded (chemically or biologically) (1)	1
08.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> • Light: UV light degrades many pesticides (eg neonicotinoids break down fastest in sunlight) so persistence is shorter in bright conditions (1) • Temperature: higher temperatures increase the rate of chemical breakdown / microbial decomposition (1) • Soil microbes / oxygen: aerobic microbial activity breaks pesticides down so persistence is shorter in well-aerated, biologically active soils (1) • pH / presence of other chemicals may increase or decrease the rate of degradation (1) • Adsorption onto clay / organic matter may protect from degradation and increase persistence (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-266; AQA spec 3.4.3.2.8.	
09.1	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Public opposition following accidents (Three Mile Island 1979, Chernobyl 1986, Fukushima 2011) (1) • Very complex, expensive technology needing an advanced industrial infrastructure (1) • Concerns about links between civil nuclear materials and weapons / terrorism (1) • Uncertainty over permanent disposal of radioactive waste (1) • Uncertain decommissioning costs (1)	3

Question	Marking guidance	Marks
09.2	<i>1 mark for each point.</i> • Only 0.7% of natural uranium is fissile U-235; the remaining 99.3% is non-fissile U-238 (a 'fertile' fuel) (1) • Neutron bombardment in the reactor converts U-238 to fissile plutonium-239 (via beta decays) (1) • So much more energy is harnessed from the mined uranium; fast reactors need no moderator but are complex and costly (1)	3
09.3	<i>2 marks for each point (partial credit for a partly correct point).</i> • Advantages: thorium is three times more abundant; much less radioactive waste with shorter half-lives; harder to make weapons material; no fuel enrichment needed (2) • Disadvantages: thorium-232 is not fissile so must be bred into uranium-233 (slow breeding rate); U-233 is an alpha emitter so very hazardous to handle; less developed technology with high development costs (2)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 192-194; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Control of greenhouse gases: energy conservation (transport, buildings, industry), renewable and nuclear energy replacing fossil fuels, CCS, reduced deforestation and afforestation (carbon sequestration), methane capture from landfill / digesters, HFC controls, changes in agriculture • Geoengineering: white roofs, ocean fertilisation, solar shades - largely untried, possible unpredictable consequences • International agreements: Kyoto Protocol, Paris Agreement - targets, but non-binding / uneven participation; carbon trading and taxes; comparison of per capita carbon footprints between countries • Evidence of limited success: global CO₂ concentrations still rising (about 280 ppm pre-industrial to over 410 ppm); temperatures up about 1 °C; emissions in some developed countries falling but rising in developing economies • Reasons for difficulty: fossil fuels underpin economies, need for major lifestyle change, cost, time delays and natural fluctuations masking effects, political disagreement, developing countries' right to develop • Adaptation strategies (flood defences, drainage, managed retreat, infrastructure changes) reduce impacts but do not address the cause • Comparison with the Montreal Protocol's success for ozone • Judgement: partial success - technology exists and some regional reductions achieved, but global strategies have not yet stopped the rise; tipping points make urgency greater	9
	Model answer - Level 3 (8 marks)	
	Strategies fall into control of greenhouse gases, geoengineering, international agreements and adaptation. Emissions are cut by energy conservation in transport, buildings and industry, by replacing fossil fuels with renewables and nuclear, by carbon capture and storage, by reducing deforestation and planting forests, and by capturing methane from landfill. Geoengineering ideas - white roofs, ocean fertilisation, solar shades - remain largely untried with unpredictable side effects. Adaptation, such as flood defences and managed retreat, reduces impacts but not the cause.	

Question	Marking guidance	Marks
	The Kyoto Protocol and Paris Agreement set targets, but they are weak or non-binding and participation is uneven; carbon trading and taxes exist but per capita footprints still differ enormously between countries. The evidence of success is limited: atmospheric CO ₂ has risen from about 280 ppm before industrialisation to over 410 ppm and temperature by about 1 °C, and although emissions are falling in some developed countries they are rising in developing economies.	
	The difficulty is that fossil fuels underpin economies, so control demands lifestyle change, cost, and political agreement between nations with different priorities, while time delays and natural fluctuations mask the effect of any action. By contrast the Montreal Protocol succeeded because ozone was a substitution problem. The judgement must be partial success: the technology exists and regional reductions prove it works, but global strategies have not yet stopped the rise, and tipping points make the delay dangerous.	
	Weaker answer - Level 2 (5 marks)	
	Methods to reduce climate change include using renewable energy such as wind and solar instead of fossil fuels, saving energy, and planting trees to absorb carbon dioxide. International agreements like the Kyoto Protocol and the Paris Agreement have set targets for countries to reduce emissions. However carbon dioxide levels are still rising, so the methods have not been very successful so far. This is because countries rely on fossil fuels and do not always keep to the agreements. More needs to be done.	
	What makes the difference	
	• Level 2 covers the main strategies and reaches a conclusion, but with no quantitative evidence (280 to 410 ppm, 1 °C) and no mention of geoengineering or adaptation.	
	• Level 3 explains <i>why</i> success is limited (economic dependence, time lags, natural variability, political disagreement) rather than just stating that it is.	
	• The Montreal comparison and the tipping-point point give Level 3 the breadth and judgement the top band requires.	
10.2	<i>1 mark.</i> • The total greenhouse gas / CO₂ emissions caused by an individual, organisation, product or country (often expressed per capita per year) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-107; AQA spec 3.2.1.2.	

Question	Marking guidance	Marks
11.1	Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Composition and structure: N₂, O₂, Ar, CO₂, ozone; troposphere and stratosphere; pressure and liquid water; gases for biological molecules and industry • Energy processes: UV absorbed by O / O₂ / O₃ dynamic equilibrium; visible light absorbed at the surface; infrared absorbed by greenhouse gases (33 °C warming); heat distribution by winds and ocean currents; transport of water vapour; energy budget and albedo • Global climate change: anthropogenic greenhouse gases, sources, feedback and tipping points, consequences for climate, sea level, currents, species; difficulties of monitoring and prediction; control strategies and agreements • Ozone depletion: CFCs and halons, catalytic destruction, effects of UV, Montreal Protocol and replacements, recovery time • Other threats: aerosols, smoke, land-use change; interconnection of processes • Evaluation: which functions are most at risk; success of ozone controls vs slow progress on greenhouse gases; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-113; AQA spec 3.2.1.	25

Question	Marking guidance	Marks
11.2	Discuss how the design of a sampling programme and the choice of statistical analysis affect the validity of conclusions drawn from environmental investigations. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Attributes: accuracy, precision, representative samples, reliability, anomalies, validity; hypothesis and null hypothesis; preliminary studies • Sampling design: random vs systematic location and avoiding bias; timing (diurnal, seasonal, long-term, weather); sample size and number and the running mean; standardisation of technique and control of other variables • Techniques for abiotic and biotic factors and their limitations (quadrats, transects, traps, mark-release-recapture assumptions, keys, meters and calibration) • Statistics: variability, standard deviation and error bars, significance and p-values, correlation vs causation; choosing Spearman's rank, chi-squared, t-test, Mann-Whitney U • Application to Paper 1 topics: monitoring pollution (noise, air quality, water), climate proxies, energy surveys, soil analysis • Evaluation of trade-offs (cost, time, disturbance) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 420-423; AQA spec 3.7.	25